

Matheus Fonseca

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EXPERIENCE

AlgoVerse

AI Researcher

Remote

Aug 2025 – Present

- Researching model merging techniques to combine complementary LLMs for improved adversarial robustness.
- Built pipeline to assess pairwise disjointness across transformer architectures and identify complementary model pairs.
- Exploring merging methods for models with similar parameter counts and layer structures.
- Working toward peer-reviewed publication in machine learning.

Cava Group, Inc.

Team Member

New York, NY

Apr 2025 – Sep 2025

- Demonstrated adaptability across multiple fast-paced roles including line service, grill, prep, and dishwashing.
- Consistently recognized by management for exceptional work ethic, curiosity, and high-quality customer service.
- Recognized for promotion to managerial position but chose to leave to prioritize studies.

PROJECTS

SendToMac | python, supabase, rumps, swift

- Developed a **cross-platform application** enabling asynchronous transfer of text and URLs from iOS to macOS, even while the Mac is offline.
- Implemented **background macOS client** using **Python**, **AppKit**, and **rumps** to display real-time alerts with user actions (Open, Snooze, Ignore).
- Prototyped iOS interface using **SwiftUI** with AI assistance to accelerate design and implementation.
- Enabled content transfer via **REST API** with responsive success/failure alerts.
- Integrated scheduling and expiration logic using timestamps to manage message flow and cleanup.

Balatest (backend) | python, flask, pypdf2, gemini, prompt engineering

- **Led a 4-person team** from ideation to completion at **CCNY's Byte Hacks**, delivering a playable educational card game prototype in 24h earning positive feedback from judges/participants.
- Engineered **backend service** in **Flask** to parse PDF content with PyPDF2 and dynamically generate JSON-based playing cards across multiple difficulty levels.
- Integrated **Gemini API** with prompt engineering techniques for semantic processing, enabling adaptive gameplay mechanics and content generation.
- Collaborated on game logic and design to align backend functionality with front-end mechanics and user experience.

Chess Web-App | html, css, javascript

- Developed an interactive **chess web application** with full rules enforcement, including check, checkmate, stalemate, and legal move validation.
- Implemented real-time piece selection with visual highlights and move restrictions based on king safety and game rules.
- Built modular move logic and game state tracking to ensure accurate gameplay flow and turn handling.
- Designed a responsive, user-friendly interface using **HTML**, **CSS**, and **JavaScript**.

EDUCATION

CUNY Brooklyn College

Bachelor of Science, Major in *Computer Science*

Expected 2029

CS50x: Introduction to Computer Science – Harvard/edX

Dec 2024

- DSA and memory management in **C**; web development in **Python**, **JavaScript**, **SQLite**, and **HTML/CSS**.

CodePath – TIP102 (Advanced Track)

Nov 2025

- Selective program covering **data structures**, **algorithms**, and problem-solving patterns through 50+ coding problems.

SKILLS

Technical: Python, Flask, C, JavaScript, CSS/HTML, SQLite, Supabase, Git, REST APIs

Spoken Languages: English (Native), Portuguese (Native), Spanish (Basic)

Interests: Chess, Weightlifting, Swimming, Travel, AI, Movies